

**Test Yourself**

**Macroeconomics**

**Circular Flow of**

**Income**

## Section A

Duration of each question is **40 mins**

1. With the help of a circular flow of income diagram, explain how equilibrium is determined in an open economy **and** consider whether a budget deficit or a trade deficit is more likely to cause lasting disequilibrium. (8) **q4a/23/M/J/23**
2. Assess whether an increase in investment in new infrastructure projects will always have a positive impact on the circular flow of income in an economy. (12) **q4b/23/M/J/23**
3. Explain the circular flow of income in a closed economy **and** consider the extent to which the circular flow of income changes as an economy becomes an open economy. (8) **q4a/22/F/M/23**
4. With the help of a diagram, explain what is meant by the circular flow of income in an open economy **and** consider the extent to which it can explain economic growth in such an economy. (8) **q4a/22/M/J/24**

## Section B

Duration **20 mins**

1. The table illustrates macroeconomic data for an economy. All figures are in \$ billions.

What is the equilibrium real output?

|          | consumption expenditure | investment | government expenditure | exports | imports | real output |
|----------|-------------------------|------------|------------------------|---------|---------|-------------|
| <b>A</b> | 110                     | 100        | 50                     | 10      | 20      | 100         |
| <b>B</b> | 120                     | 100        | 60                     | 20      | 30      | 200         |
| <b>C</b> | 140                     | 100        | 70                     | 30      | 40      | 300         |
| <b>D</b> | 160                     | 100        | 80                     | 40      | 50      | 430         |

2. The table lists the values of the components of an economy's circular flow of income.

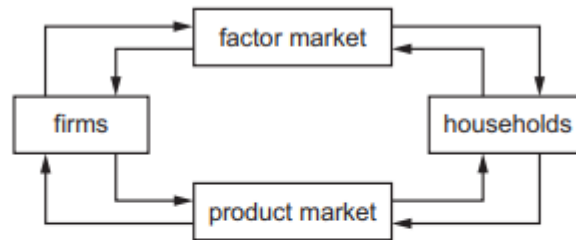
| component           | value \$m |
|---------------------|-----------|
| government spending | 6         |
| exports             | 8         |
| investment          | 9         |
| imports             | 7         |
| saving              | 10        |
| tax                 | 5         |

What can be concluded about the economy from the information shown?

- A** It has a budget surplus.
- B** It has a trade deficit.
- C** It is a mixed, open economy.
- D** Its circular flow is in equilibrium.

3. What is most likely to increase a country's circular flow of income?
- A** Its budget deficit increases.
  - B** Its imports increase.
  - C** Its interest rates increase.
  - D** Its exchange rate increases.
4. In the circular flow of income model of an economy, Y, C, I, G, X, S, T and M represent total income, consumption, investment, government expenditure, exports, saving, taxation and imports respectively.
- Which statement is correct?
- A** At equilibrium, the levels of Y and C are always the same.
  - B** Governments always adjust G so that the economy is in equilibrium.
  - C** If I is not equal to S, it is still possible for the national income to be in equilibrium.
  - D** If the national income is in equilibrium, X must be equal to M.
5. Which statement about the circular flow of income is correct?
- A** Exports and imports do not affect the equilibrium level of national income because they always cancel each other out.
  - B** The greater the level of saving, the greater the level of injections in the form of investment must be.
  - C** When consumers use past savings to increase their spending, there will be a rise in national income.
  - D** When a government increases withdrawals by increasing the rate of income tax, there must be an equivalent increase in injections to compensate.

6. The diagram shows a basic two-sector circular flow of income.



What can be concluded about the direction of flows?

- A Consumer expenditure flows anticlockwise.
  - B Goods and services flow clockwise.
  - C Incomes flow anticlockwise.
  - D Resources flow anticlockwise.
7. General equilibrium in the macroeconomy is achieved when total injections are equal to total leakages.

What is most likely to cause disequilibrium in the macroeconomy in the short run?

- A an increase in government spending of \$10bn and a decrease in exports of \$10bn
  - B an increase in investment of \$30bn and a decrease in household savings of \$30bn
  - C an increase in spending on imports of \$40bn and an increase in government spending of \$40bn
  - D an increase in taxation of \$40bn and a decrease in spending on imports of \$40bn
8. Which changes will increase the circular flow of income?

|   | higher imports | higher investment | higher taxation |
|---|----------------|-------------------|-----------------|
| A | yes            | no                | no              |
| B | yes            | yes               | no              |
| C | no             | yes               | yes             |
| D | no             | yes               | no              |

9. Aggregate demand in an economy may decrease as a result of an increase in
- A consumption expenditure.
  - B government expenditure.
  - C import expenditure.
  - D investment expenditure.

# MARKING SCHEME

## SECTION A

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1        | <p><b>With the help of a circular flow of income diagram, explain how equilibrium is determined in an open economy and consider whether a budget deficit or a trade deficit is more likely to cause lasting disequilibrium.</b></p> <p>Follow the point-based marking guidance at the top of this mark scheme and award:<br/> up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis<br/> up to 2 marks for AO3 Evaluation.</p> <p><b>AO1 Knowledge and understanding (max 3 marks)</b><br/> An understanding of the circular flow of income shown through an accurate diagram showing income flows between firms and households (1); correct inclusion of exports and imports (1); correct inclusion of government spending and taxation (1)</p> <p><b>AO2 Analysis (max 3 marks)</b><br/> Explanation of injections into the circular flow of income;(1); explanation of leakages from the circular flow of income (1); determination of equilibrium in the circular flow of income (1)</p> <p><b>AO3 Evaluation (max 2 marks)</b><br/> Consideration how a budget deficit or a trade deficit leads to disequilibrium in the circular flow of income (1); consideration which is likely to cause lasting disequilibrium (1) <b>Reserve 1 mark for a justified conclusion</b></p> | 8     |
|          | AO1 Knowledge and understanding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3     |
|          | AO2 Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |
|          | AO3 Evaluation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2     |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2        | <p><b>Assess whether an increase in investment in new infrastructure projects will always have a positive impact on the circular flow of income in an economy.</b></p> <p>Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question.</p> <p>AO1 and AO2 out of 8 marks. AO3 out of 4 marks.</p> <p><b>Indicative content</b></p> <p>Responses may include:</p> <p><b>AO1 Knowledge and understanding and AO2 Analysis (max 8 marks)</b><br/>Increased investment in new infrastructure can have a positive and negative impact on the circular flow of income in an economy:</p> <ul style="list-style-type: none"> <li>• Increased investment can be from firms or government</li> <li>• Household savings can be a source of funds for this investment</li> <li>• Investment is an injection in to the circular flow of income</li> <li>• There is increased payments for factors of production</li> <li>• Households supply factor services to firms in the form of labour, enterprise and capital</li> <li>• In turn, firms provide factor services to implement the investment</li> <li>• Leakages impact on the total flow of funds in the circular flow of income</li> <li>• Taxation on the inputs into the new infrastructure project is a leakage from households and firms to the government</li> <li>• Other leakages are spending on imports of materials to implement the new infrastructure project</li> <li>• Household savings are a leakage from factor income</li> <li>• Equilibrium in the circular flow will depend on the balance between injections and leakages.</li> </ul> <p><b>Maximum level 2 for a one-sided response</b></p> <p><b>AO3 Evaluation (max 4 marks)</b></p> <ul style="list-style-type: none"> <li>• The circular flow model is a simplified view of reality.</li> <li>• The impact will depend upon the net of injections and leakages.</li> <li>• There could be a negative impact on the circular flow of income if leakages exceed injections.</li> <li>• Most forms of new infrastructure have a negative impact not taken into account by the model</li> <li>• Consideration as to whether all new infrastructure always has a positive impact on the circular flow of income</li> </ul> <p>Accept all valid responses.</p> <p><b>A one-sided response cannot gain any marks for evaluation.</b></p> | 12    |
|          | AO1 Knowledge and understanding and AO2 Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8     |
|          | AO3 Evaluation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3        | <p><b>Explain the circular flow of income in a closed economy and consider the extent to which the circular flow of income changes as an economy becomes an open economy.</b></p> <p>Follow the point-based marking guidance at the top of this mark scheme and award:<br/> up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis<br/> up to 2 marks for AO3 Evaluation.</p> <p><b>AO1 Knowledge and understanding (max 3 marks)</b></p> <p>A clear knowledge and understanding that the circular flow of income in a closed economy involves households, firms, and the government (1) and has injections of I and G and leakages/withdrawals of S and T (1), so that equilibrium income is where <math>I + G = S + T</math> (1).</p> <p><b>AO2 Analysis (max 3 marks)</b></p> <p>An analysis of how the circular flow of income changes as an economy becomes an open economy, with the introduction of participation in the international economy (1) through the injection of X and the leakage/withdrawal of M (1), so that equilibrium income (is) (now becomes) where <math>I + G + X = S + T + M</math> (1).</p> <p><b>AO3 Evaluation (max 2 marks)</b></p> <p>Offers a valid judgement on the extent to which the circular flow of income changes as an economy becomes an open economy depending on the importance of international net trade <b>(1)</b> to reach a conclusion. <b>(1)</b></p> | 8     |
|          | AO1 Knowledge and understanding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3     |
|          | AO2 Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3     |
|          | AO3 Evaluation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4        | <p><b>With the help of a diagram, explain what is meant by the circular flow of income in an open economy <u>and</u> consider the extent to which it can explain economic growth in such an economy.</b></p> <p>Follow the point-based marking guidance at the top of this mark scheme and award:<br/> up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis<br/> up to 2 marks for AO3 Evaluation.</p> <p><b>AO1 Knowledge and Understanding (max 3 marks)</b><br/> A simple circular flow diagram (1) that clearly shows/ explains separately that expenditure flows from households to firms (1) and that income flows from firms to households (1) <b>Note:, this is all that is required for the knowledge and understanding marks regardless of the complexity of the diagram.</b></p> <p><b>AO2 Analysis (max 3 marks)</b></p> <ul style="list-style-type: none"> <li>The first two marks are for an analysis of injections greater than withdrawals and how this leads to economic growth/increase in national income and (1) for an analysis of withdrawals greater than injections and how this leads to a fall in economic growth/national income (1).</li> <li>Reference must be made in some way to the fact that in open economy injections include income from exports and that leakages includes expenditure on imports along with at least one other example of an injection and one other example of a leakage for the (1).</li> </ul> <p><b>AO3 Evaluation (max 2 marks)</b></p> <ul style="list-style-type: none"> <li>Evaluation should explain how the relative sizes of injections and leakages will lead either to economic growth or a fall in economic growth with 1 mark reserved for a justified conclusion.</li> </ul> <p><b>Please use a text box to show the mark split e.g., 2,2,1</b></p> | 8     |
|          | AO1 Knowledge and understanding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3     |
|          | AO2 Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3     |
|          | AO3 Evaluation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |

## **SECTION B**

**1. C**

**2. C**

**3. A**

**4. C**

**5. C**

**6. D**

**7. B**

**8. D**

**9. C**